Utilizing Spatial Modeling and Socioeconomic Indicators to Optimize Public Health Resource Allocation in Boston

Sarah Yeh '27 Data Science Major Capstone

Introduction

Research Question

Which geographical areas (Census tracts) and types of food establishments in Boston exhibit the highest frequency and severity of health code violations between 2006 and 2026? How can predictive modeling of these patterns inform inspection resource allocation and identify establishments with the highest potential for intervention and improvement?

Background

Observation suggests that health code violations are disproportionately clustered in certain Boston census tracts. Understanding these geographical disparities is critical for optimizing the city's limited public health resource.

Data

Sources

- Boston Food Inspection Data (2006-2026)
- US Census Tract Data (TIGER/Line Shapefiles)

Data Cleaning/Manipulation

- Handled missing data in ZIP codes, violation levels.
  - Only included violation levels: *, **, ***.
- Only included stationary restaurants, **not** mobile food establishments.
- Created predictor variables: Month of inspection
- Standardized ZIP codes, merged inspection data with the TIGER/Line shapefiles at the Census Tract level and Census Tract data for income level.

Visualizations

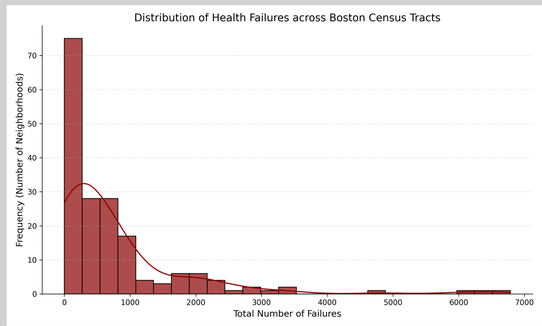

Figure 1. Distribution of health failures across Boston Census Tracts.

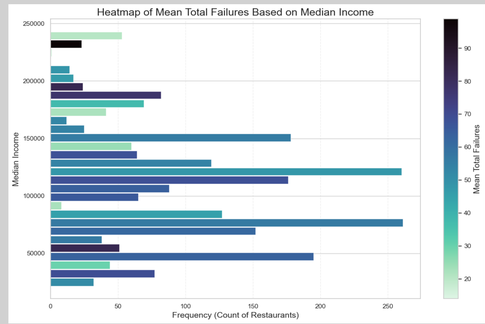

Figure 2. Heatmap of mean total failures based on median income.

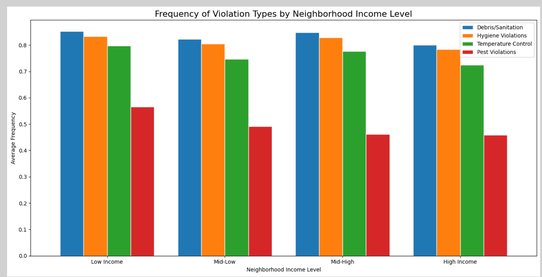

Figure 3. The frequency of violation types by neighborhood income level (debris, hygiene, temperature, and pest violations.

Figure 4. Geographical heatmap of inspection failures across Boston.

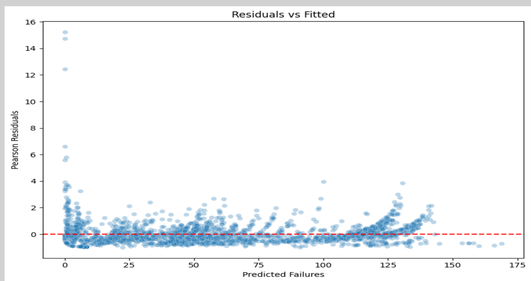

Figure 5. *Residual vs. Fitted Plot of the Negative Binomial Model.*

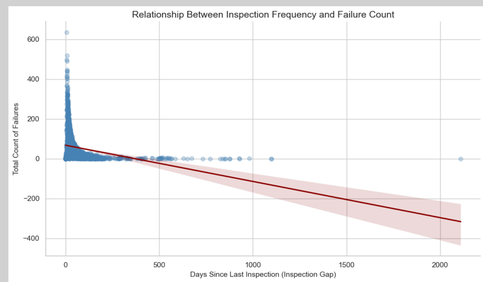

Figure 6. *Relationship between Inspection Frequency and Failure Count.*

Methods/Results

Method

- Data Integration: Merged 20 years of Boston Food Inspection data with Census Tract shapefiles using a spatial join. Missing spatial data (<5%) was excluded.
- Model Specification: Utilized a **Negative Binomial Generalized Linear Model**.
- Feature Engineering: Violation Types (Debris, Hygiene, and Pests), is_chain, income-pest interaction.
- To address multicollinearity, used VIF analysis showing that there are not variables that violate multicollinearity.
- Compared base model (without neighborhood level variables) and full model (including neighborhood variables)
  - Train/Test split: Model was trained on data from 2006-2024 and validated using modern data from 2025-2026.

Model	AIC	Pseudo R2
Base Model	**20519.14**	0.7469
Full Model	20523.64	0.7471

Results

- After comparing base model and full model using the Likelihood Ratio Test with (**p = 0.68**) and AIC analysis, it is revealed that neighborhood demographics **do not** provide predictive power beyond operational flags.
- Used **base model** to predict total_failures in restaurants.
- Train data resulted in a Pseudo R2 of **0.7133**.
- Predictive Power: Test data resulted in a Pseudo R2 of **0.5461**

Variable	Coeff (log)	P-value
Flag_hygiene	1.1820	< 0.001
Flag_pests	0.7741	< 0.001
Flag_debris	1.2928	< 0.001
Is_chain	-0.1494	0.385
inspection_gap	-0.0072	< 0.001
year	-0.0924	< 0.001

Conclusion and Limitations

Discussion/Conclusion

Independent establishments and specific high-traffic Census Tracts show higher raw violation counts. However, modeling reveals that this results from operational behavior with violations being from debris, hygiene, and pests rather than neighborhood status (p = 0.68). There appears to be a negative correlation between inspection gaps and failures, proving that restaurants with more fails have more inspection visits. When trained on the test data, the Negative Binomial model explained **71.55%** of the variance, showing that risk is highly predictable when solely focusing on the internal management of the restaurant. Ultimately, predictive modeling identifies that kitchen behavior **outweighs** neighborhood environment.

Limitations

Some limitations may include a reporter and observer bias as there is a potential for higher complain rates in wealthy tracts and increased inspections in lower-income areas. Additionally, the dataset lacks variables such as staff turnover rates, which may contribute to the higher failure rates in inspections. The analysis underwent some loss of data when data cleaning and merging Census Tract data with the inspection data, which may result in an under representative sample of volatile neighborhoods.